AKAP11

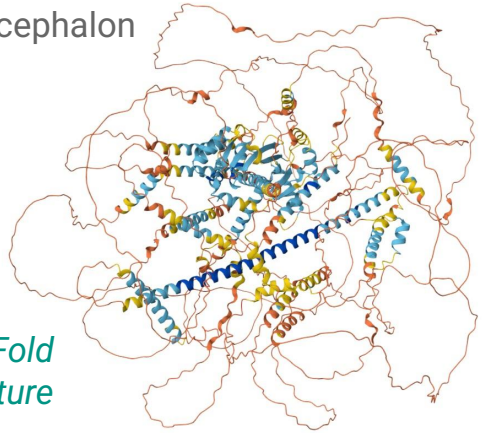
by Jas

Background

- A-kinase anchoring protein 11
- protein-coding (codes AKAP11)
- AKAP family (cAMP)
- schizophrenia risk gene

expressed in *(from Allen Brain Atlas)*:

- diencephalon (less)
- **mesencephalon**
- somewhat in between at cerebellum
- **pons**
- bleeds over to myelencephalon



AlphaFold experimental structure

Knockout Mice

- https://www.modelorg.us/portal/article/index/id/10909/post_type/3.html
- https://www.cyagen.com/us/en/catalog-model-bank-us/S-KO-05497

- still in development, none currently available (for now)

Akap11-KO

Nomenclature	C57BL/6Smoc-*Akap11*^{em1Smoc}
Cat. NO.	TBD
Strain	Developing
State ⓘ	

ALGGEN - PROMO

Factors predicted within a dissimilarity margin less or equal than 7 % :

0 c-Jun [T00133] 1 GCF [T00320] 2 Pax-5 [T00070] 3 p53 [T00671]
4 E2F-1 [T01542] 5 ER-alpha [T00261] 6 RXR-alpha [T01345] 7 MAZ [T00490]
8 Ik-1 [T02702] 9 Sp1 [T00759] 10 WT1 [T00899] 11 Egr-1 [T00241]
12 STAT4 [T01577] 13 c-Ets-1 [T00112] 14 Elk-1 [T00250]

Zoom Data (txt)

Distribution of the nucleotides over the given chain:

Your Input:

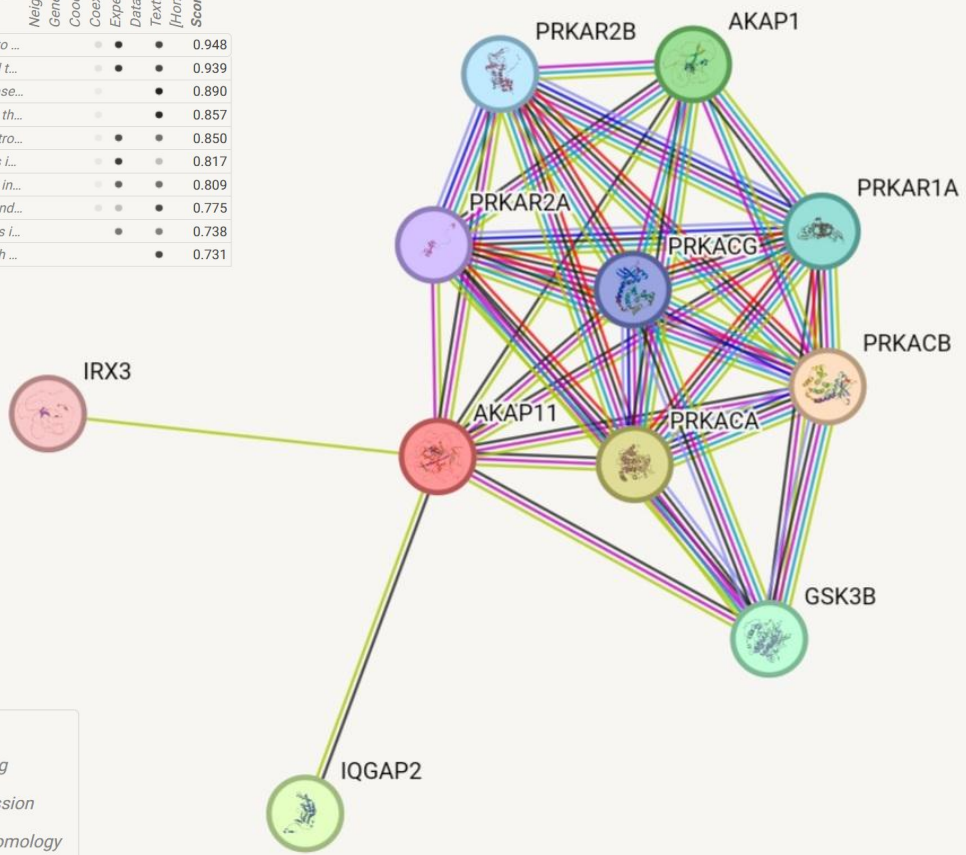
		Neighborhood	Gene Fusion	Cooccurrence	Coexpression	Experiments	Databases	Textmining	[Homology]	Score
AKAP11	*A-kinase anchor protein 11; Binds to type II regulatory subunits of protein kinase A and anchors/targets them; Belongs to the AKAP110 family. (1901 aa)*									

Predicted Functional Partners:

PRKACB	*cAMP-dependent protein kinase catalytic subunit beta; Mediates cAMP-dependent signaling triggered by receptor binding to ...*						●		●	0.948
PRKACA	*cAMP-dependent protein kinase catalytic subunit alpha; Phosphorylates a large number of substrates in the cytoplasm and t...*						●		●	0.939
IQGAP2	*Ras GTPase-activating-like protein IQGAP2; Binds to activated CDC42 and RAC1 but does not seem to stimulate their GTPase...*								●	0.890
AKAP1	*A-kinase anchor protein 1, mitochondrial; Binds to type I and II regulatory subunits of protein kinase A and anchors them to th...*								●	0.857
GSK3B	*Glycogen synthase kinase-3 beta; Constitutively active protein kinase that acts as a negative regulator in the hormonal contro...*						●		●	0.850
PRKAR1A	*cAMP-dependent protein kinase type I-alpha regulatory subunit; Regulatory subunit of the cAMP-dependent protein kinases i...*						●		●	0.817
PRKAR2B	*cAMP-dependent protein kinase type II-beta regulatory subunit; Regulatory subunit of the cAMP-dependent protein kinases in...*						●		●	0.809
PRKACG	*cAMP-dependent protein kinase catalytic subunit gamma; Phosphorylates a large number of substrates in the cytoplasm and...*						●		●	0.775
PRKAR2A	*cAMP-dependent protein kinase type II-alpha regulatory subunit; Regulatory subunit of the cAMP-dependent protein kinases i...*						●		●	0.738
IRX3	*Iroquois-class homeodomain protein IRX-3; Transcription factor involved in SHH-dependent neural patterning. Together with ...*								●	0.731

- mostly co-expression + textmining
- AKAP11 bonds: purely co-expression, textmining, experimentally determined
- interwoven: gene fusions, gene co-occurrence, protein homology
- overall seems unsure

- most cAMP-related or dependent proteins



Known Interactions	Predicted Interactions	Others
from curated databases	*gene neighborhood*	*textmining*
experimentally determined	*gene fusions*	*co-expression*
	gene co-occurrence	*protein homology*

Name: Untitled

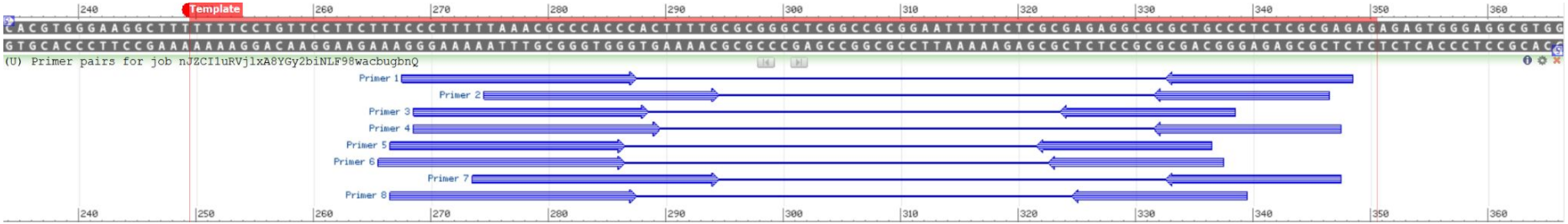
Conformation: linear

Enzymes: BstUI

Noncutters:

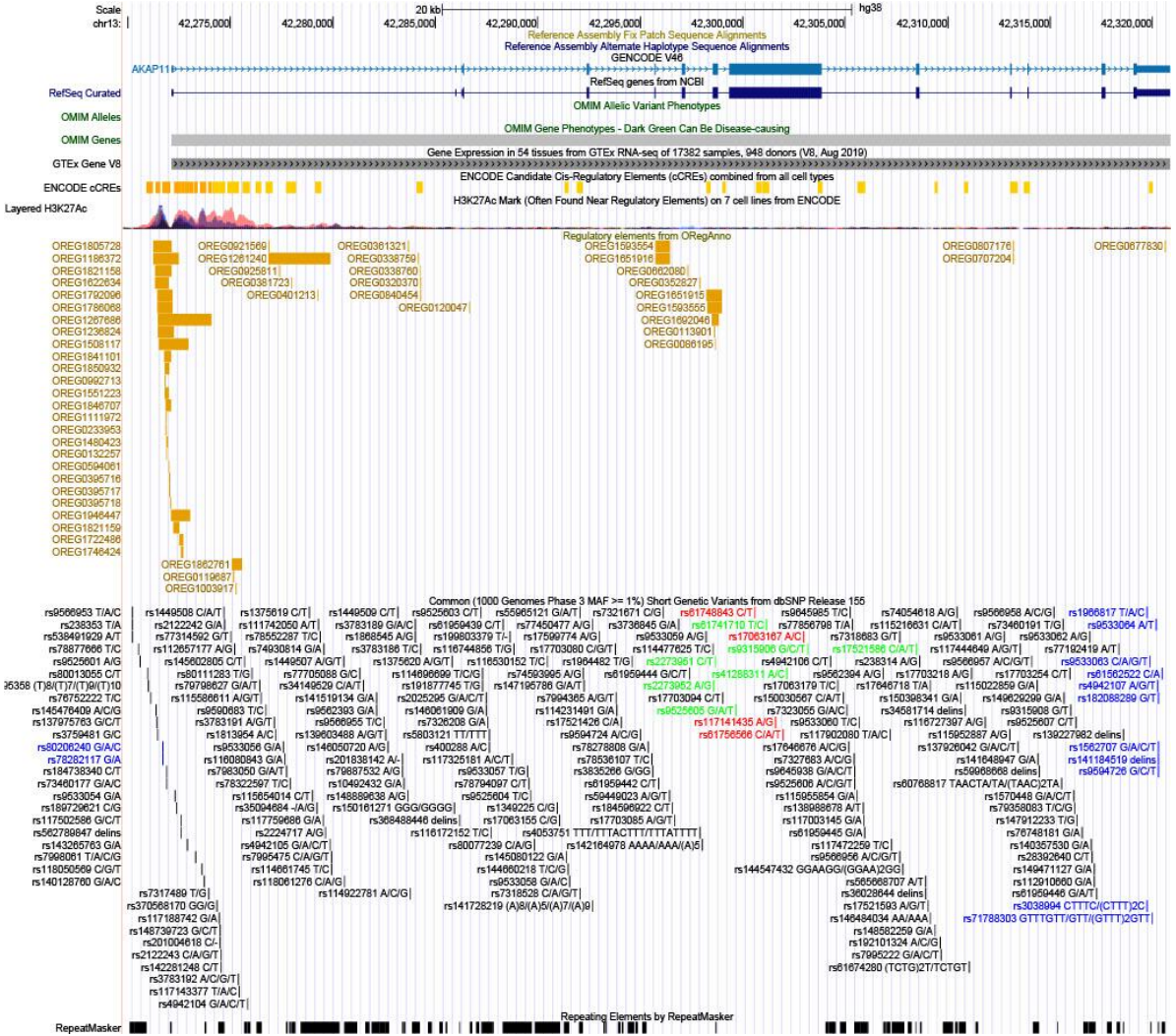
https://restrictionmapper.org/

Name	Sequence	Site Length	Overhang	Frequency	Cut Positions
BstUI	CGCG	4	blunt	10	88, 206, 218, 296, 308, 322, 331, 344, 449, 482



NIH Primer-BLAST:

https://www.ncbi.nlm.nih.gov/tools/primer-blast/primertool.cgi?ctg_time=1722479772&job_key=nJZCI1uRVjlxA8YGy2biNLF98wacbugbnQ



UCSC Genome Browser

promoter SNP:
e.g. rs238353 T/A
(possibly no active function as compared to others?)

- upstream transcript variant, no reported clinical significance

coding SNP:
e.g. rs61748843 C/T
(protein-altering variants or splice site variants)

- missense variant, benign
- oddly large amount of UTR variants near **end** of gene (terminator)
- most functional variants (protein-altering, splice site, or synonymous codon variants) in coding found within or around **exon 8**

Schizophrenia + Bipolar Disorder

- risk gene in both schizophrenia and bipolar disorder
- seem to be heavily related
- AKAP11 deficiency as risk (besides protein alteration in both schizophrenia and bipolar)

schizophrenia:

- mutant mice study: AKAP11 deficiency/mutant EEG scans parallel human schizophrenia (GRIN2A also)

bipolar:

- AKAP11 interacts with GSK3B (hypothesized to be targeted by lithium)
  - lithium as existing BD treatment–AKAP11 or GSK3B possible drug targets?

- still other possible factors…other genes…
- neurological illnesses very complicated

Bibliography

- https://pubmed.ncbi.nlm.nih.gov/36914870/
- https://human.brain-map.org/microarray/search/show?exact_match=false&search_term=%22AKAP11%22&search_type=gene (Allen Brain Atlas)
  - https://www.ncbi.nlm.nih.gov/books/NBK551509/
  - https://www.ncbi.nlm.nih.gov/books/NBK234157/
- https://alphafold.ebi.ac.uk/entry/Q9UKA4 (AlphaFold)
- https://alggen.lsi.upc.es/cgi-bin/promo_v3/promo/promoinit.cgi?dirDB=TF_8.3 (ALGGEN - PROMO)
- https://string-db.org/cgi/network?taskId=bNSGCkbOy2zl&sessionId=bbq44gFyXkS (STRING)
- Eukaryotic Promoter Database–three possibilities (used **AKAP11_1**)
  - https://epd.expasy.org/cgi-bin/epd/get_doc?db=hgEpdNew&format=genome&entry=AKAP11_1
- primers:
  - https://restrictionmapper.org/ (to find cut sites)
  - https://www.ncbi.nlm.nih.gov/tools/primer-blast/primertool.cgi?ctg_time=1722479772&job_key=nJZCI1uRVjlxA8YGy2biNLF98wacbugbnQ (actual primer results)

Bibliography (cont.)

- https://genome.ucsc.edu/cgi-bin/hgTracks?db=hg38&lastVirtModeType=default&lastVirtModeExtraState=&virtModeType=default&virtMode=0&nonVirtPosition=&position=chr13%3A42269754%2D42320863&hgsid=2321212104_xLa0FhlxvTKtRaTCyUbX0AoFr0Yx (USCS Genome Browser)
  - https://www.ncbi.nlm.nih.gov/snp/rs238353
  - https://www.ncbi.nlm.nih.gov/snp/rs61748843
- https://pubmed.ncbi.nlm.nih.gov/35410376/
- https://pubmed.ncbi.nlm.nih.gov/36914641/
- https://pubmed.ncbi.nlm.nih.gov/37171958/

End

Thank you!